Ridgway's Rail Passive Acoustic Monitoring

A manager's report for designing future scientific studies

Sound Ecology Pilot Studies

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1 Executive summary

This report translates the Ridgway's Rail section of the Sound Ecology Pilot Studies repository into a decision aid for managers. It describes what passive acoustic monitoring (PAM) can contribute to future studies, what the pilot demonstrated, and which design decisions should be resolved before committing to a larger survey.

1.1 Decisions this report supports

Managers can use this report to:

1. decide whether PAM is appropriate for a monitoring objective;
2. select a recording and analysis design that matches that objective;
3. distinguish pilot evidence from results that require validation; and
4. plan a defensible, reproducible field study.

The pilot is best treated as a methods study, not as a final population estimate. Detection probability, call classification, habitat effects, and availability of independent detections all need to be quantified before inference is extended beyond the sampled locations.

1.2 How to use the report

The chapters provide a concise management narrative. The final chapter is a crosswalk to the complete interactive components in the repository, including the dashboard, methods, population assessment, biodiversity, insects, bats, sound exploration, and future-research pages. Those pages retain the detailed figures, code, and exploratory results.

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2 Study design and monitoring objectives

2.1 Start with the management question

PAM is most useful when the management question is explicit. A study intended to map occupancy has different requirements from one intended to estimate density, evaluate habitat, or measure response to restoration. Define the estimand, the time period, the spatial frame, and the decision threshold before selecting recorder settings.

2.2 What the pilot tested

The Ridgway's Rail pilot combined deployments at the Tijuana Estuary and Sweetwater Proving Pens with several recording schedules. Continuous 48-kHz recording supported temporal analyses; sunrise/sunset sampling extended coverage over a longer period; and high-frequency overnight sampling supported bat detection. This contrast is valuable for planning, because it makes the trade-off between temporal coverage, storage, battery life, and taxonomic coverage explicit.

2.3 Recommended design sequence

1. Define the target population and the inference area.
2. Pilot recorder placement, calibration, and detection range in representative habitat.
3. Collect a validation set of manually reviewed calls.
4. Specify the sampling schedule and replication needed for the estimand.
5. Pre-register quality-control rules, classifier thresholds, and exclusion criteria.

Spatial and temporal replication should be planned together. Sites should represent the habitat and management conditions to which results will be generalized, while repeated recordings provide information about imperfect detection.

3 Analysis and strength of evidence

3.1 Detection and classification

The pilot used BirdNET-assisted screening and manual review to identify candidate rail vocalizations. Classifier confidence is a screening measure, not a prevalence estimate. Future work should estimate sensitivity and specificity using a validation set that spans sites, recorder configurations, background noise, and call types.

3.2 Population assessment

Detection-range work and distance-based approaches are promising routes to abundance or density estimation. They are not interchangeable: point-transect detection functions require suitable distance information, while acoustic spatial capture-recapture requires identifiable detections and a model appropriate to the sensor array. A future study should choose one estimand and design the field data around its assumptions.

3.3 Community and soundscape indicators

Biodiversity, acoustic complexity, acoustic diversity, bioacoustic, and entropy indices can provide context about wetland communities and habitat condition. These indices should be treated as supporting indicators until their relationship to management outcomes is calibrated. They should not replace species-level validation when the decision concerns Ridgway's Rails.

3.4 Evidence limits

The pilot results reveal operational patterns and useful design questions. They do not, by themselves, establish population trend, causal effects of management, or transferability to unsampled wetlands. Report uncertainty and validation performance alongside every future estimate.

4 Recommendations for managers

4.1 Near-term actions

- Agree on the management objective and estimand before deployment.
- Maintain a small, manually reviewed reference library for every survey.
- Measure recorder performance and detection range in each representative habitat type.
- Keep raw audio, classifier outputs, review decisions, and metadata together.
- Use the same versioned analysis settings across sites and survey periods.

4.2 Study design priorities

Use a staged design: an initial feasibility deployment, a validation study, and then a powered monitoring study. Include repeated visits or recording windows so that non-detection can be separated from absence. If response to restoration is the objective, include reference and treatment conditions and sample before and after management.

4.3 Governance and reporting

Archive data and code with persistent identifiers, document deviations from the protocol, and publish assumptions in a form that can be audited. A managerial decision should be traceable from the question, through the sampling design and model, to the evidence and its uncertainty.

5 Implementation checklist

Before field deployment, confirm:

- ☐ The target population, estimand, inference area, and decision rule are written down.
- ☐ Recorder placement, schedule, sample rate, and replication are justified.
- ☐ Calibration and detection-range protocols are complete.
- ☐ A representative, manually reviewed validation set is available.
- ☐ Classifier thresholds and quality-control rules are version controlled.
- ☐ Storage, battery, metadata, and chain-of-custody procedures are tested.
- ☐ The analysis plan identifies how imperfect detection will be handled.
- ☐ Raw recordings and derived data will be archived with the code.
- ☐ Results will report uncertainty and limitations to decision makers.

This checklist is intentionally conservative: resolving these items costs less than discovering after deployment that a key inference cannot be supported.

6 Crosswalk to the complete Rails section

The following pages are the complete source components summarized in this report. They remain the authoritative location for interactive figures, exploratory analyses, and implementation details.

Component	Purpose
Dashboard	Deployment, recording, and rail-detection overview
Introduction	Scope and motivation for the pilot
Methods	Study area, recorders, detection, distance, and index methods
Summary	Temporal and diurnal results
Population assessment	Detection range, density, ASCR, and habitat questions
Assessment details	Supporting assessment notes and exploratory analysis
Biodiversity	Community and acoustic-index analyses
Bats	High-frequency monitoring component
Insects	Insect monitoring component
Future research	Survey-design and management recommendations
Sound exploration	Examples of recorded sounds and review context
Rails sound examples	Shared sound-analysis material used by the Rails pages

The source files for these pages are in `content/`. Data and analysis scripts are retained in `data/` and `_code/` so that future studies can reproduce or extend the workflow.